Observations

Lab note for Negative Mine hard examples.

Embedding training was ignoring near-miss passages that shared vocabulary.

The observations follow.

Negative Mine adds same-domain passages with different numbers as explicit hard negatives.

Adding those negatives lifted recall at three on paraphrased questions by four points.

The working decision is: Train with one checked hard negative per paraphrased query and discard negatives that contain the answer..

Measurement

How it was measured: Recall at three was measured on a paraphrased question set before and after one epoch with hard negatives.

What came back: Recall at three moved from 0.64 to 0.68 on the paraphrased set.

What this does not show: The mined negatives were written by hand, so the gain may shrink on automatically mined sets.

Still unresolved: How many hard negatives per query are useful before the batch becomes mostly near misses? Operational risks: Severity medium: A hard negative that is actually relevant will push the model away from a correct page.

Recommended action: Drop any negative that contains the answer sentence before training..